

Theorem and Proof

Definitions. Let $k, x, x_{eq} \in \mathbb{R}$. We define the force function $F(x)$ and the energy function $E(x)$ as follows:

$$F(x) = -k(x - x_{eq})$$
$$E(x) = \frac{k}{2}(x - x_{eq})^2$$

Theorem. The derivative of the energy function with respect to x is equal to the negative of the force function. That is,

$$\frac{d}{dx}E(x) = -F(x)$$

Proof. We begin by expanding the definition of the energy function on the left-hand side:

$$\frac{d}{dx}E(x) = \frac{d}{dx} \left[\frac{k}{2}(x - x_{eq})^2 \right]$$

By the constant multiple rule for differentiation, we can factor out the constant $\frac{k}{2}$:

$$\frac{d}{dx}E(x) = \frac{k}{2} \cdot \frac{d}{dx} [(x - x_{eq})^2]$$

Next, we apply the chain rule and the power rule to differentiate the squared term. The outer function is u^2 and the inner function is $u = x - x_{eq}$.

$$\frac{d}{dx} [(x - x_{eq})^2] = 2(x - x_{eq}) \cdot \frac{d}{dx}(x - x_{eq})$$

Since the derivative of x is 1 and the derivative of the constant x_{eq} is 0, the inner derivative is simply 1. Substituting this back yields:

$$\frac{d}{dx}E(x) = \frac{k}{2} \cdot 2(x - x_{eq}) \cdot 1$$

Simplifying the algebraic expression by canceling the 2s, we get:

$$\frac{d}{dx}E(x) = k(x - x_{eq})$$

Now, we unfold the definition of the negative force function, $-F(x)$:

$$-F(x) = -(-k(x - x_{eq})) = k(x - x_{eq})$$

Since both $\frac{d}{dx}E(x)$ and $-F(x)$ simplify to the exact same algebraic expression, we conclude that:

$$\frac{d}{dx}E(x) = -F(x)$$

This completes the proof. □